

Body Control System - High Level Design

Prototype Evaluation Document - Synthetic Data

Field	Value
Document ID	HLD-BCM-001
Version	1.0
System	Body Control System
Classification	Synthetic / Evaluation Only

1. Purpose

This High Level Design describes the software architecture of the Body Control System. The document defines software components, ports, interfaces, signals, dependencies and high-level functional flow for prototype evaluation.

2. Software Components

Component	Responsibility
LightSwitch	Provides lighting requests.
BodyControlManager	Coordinates body-electronics functions.
ExteriorLightController	Controls exterior vehicle lights.

3. Interfaces

Interface	Provider	Consumer
LightRequestInterface	LightSwitch	BodyControlManager
ExteriorLightInterface	BodyControlManager	ExteriorLightController

4. Port Definitions

Component	Port	Port Type	Interface
LightSwitch	LightRequestOut	P-Port	LightRequestInterface
BodyControlManager	LightRequestIn	R-Port	LightRequestInterface
BodyControlManager	ExteriorLightOut	P-Port	ExteriorLightInterface
ExteriorLightController	ExteriorLightIn	R-Port	ExteriorLightInterface

5. Signal Definitions

Signal	Interface	Unit
LightRequest	LightRequestInterface	enum
ExteriorLightCommand	ExteriorLightInterface	enum

6. Component Dependencies

Dependent Component	Required Component
BodyControlManager	LightSwitch
ExteriorLightController	BodyControlManager

7. Functional Flow

LightSwitch -> BodyControlManager -> ExteriorLightController

8. Revision Information

Current document version: 1.0. Document identifier: HLD-BCM-001.